

Production Readiness Report (grpc-eio)

Date: 2026-01-17

Scope: RFC 7540/7541 + gRPC core + gRPC-Web (gateway path for client streaming/bidi).

Environment / Versions

- OS: Darwin 25.2.0 (arm64)
- OCaml: 5.4.0
- Dune: 3.20.2
- ghz: 0.121.0
- Go: 1.24.1 (grpc-go interop client: google.golang.org/grpc@v1.78.0)
- Rust: 1.92.0

Summary

- Status: READY (interop + observability + performance baseline + 24h soak)
- Note: browser direct gRPC-Web supports unary/server-streaming only; client-streaming/bidi requires WebSocket gateway.

1) Interop Suite (Go/Rust clients vs OCaml server)

Run:

```
./bench/run_interop_suite.sh --repeat 1
```

Result: PASS (no FAIL/ERROR)

Log:

- tmp/interop_suite_fix_20260117_015547.log

2) Official gRPC Interop (grpc-go client)

Plaintext:

```
./bench/run_official_interop.sh
```

Result: PASS

Log:

- tmp/interop_official_fix_plain_20260117_015500/run.log

TLS:

```
USE_TLS=1 CERT_FILE=tmp_tls/server.pem KEY_FILE=tmp_tls/server.key \  
CA_FILE=tmp_tls/ca.pem SERVER_NAME=localhost \  
./bench/run_official_interop.sh
```

Result: PASS

Log:

- tmp/interop_official_fix_tls_20260117_015523/run.log

Skipped (unsupported by design):

- custom_metadata, status_code_and_message, special_status_message
- orca_per_rpc, orca_oob, pick_first_unary
- compute_engine_creds, service_account_creds, jwt_token_creds, per_rpc_creds
- oauth2_auth_token, google_default_credentials, compute_engine_channel_credentials

3) Observability Verification (Prometheus)

Health:

- Target status: up
- tmp/prom_targets_verify.json

Signal:

- grpc_server_calls_total=1
- tmp/prom_query_calls_verify.json

Rules:

- GrpcHighErrorRate, GrpcHighLatencyP95, GrpcNoTraffic
- ops/prometheus/alerts.yml

Screenshots:

- docs/assets/prom_targets.png
- docs/assets/prom_alerts.png
- docs/assets/prom_query_calls.png
- docs/assets/ghz_results.png
- docs/assets/interop_results.png

Prometheus

QueryAlertsStatus

Filter by rule group state

Filter by rule name or labels

grpc_eio /etc/prometheus/alerts.yml

INACTIVE (3)

GrpcHighErrorRate

(sum(rate(grpc_server_calls_failed_total[5m])) / sum(rate(grpc_server_calls_total[5m]))) > 0.01

for: 5m

severity="page"

description

Error ratio exceeded 1% over 5m.

summary

gRPC error rate > 1%

GrpcHighLatencyP95

histogram_quantile(0.95, sum by (le) (rate(grpc_server_call_duration_seconds_bucket[5m]))) > 0.5

for: 10m

severity="warn"

description

p95 latency above 500ms over 10m.

summary

gRPC p95 latency > 500ms

GrpcNoTraffic

sum(rate(grpc_server_calls_total[5m])) == 0

for: 10m

severity="warn"

description

No calls observed in the last 10m.

summary

gRPC traffic halted

 Prometheus

Query

Alerts

Status

Filter by rule group state

Filter by rule name or labels

grpc_eio /etc/prometheus/alerts.yml

INACTIVE (3)

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Status ▾

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>_ grpc_server_calls_total

:

Execute

Table

Graph

Explain

< Evaluation time >

Load time: 11ms Result series: 2

grpc_server_calls_total{instance="host.docker.internal:9464", job="grpc_eio"}1

grpc_server_calls_total{instance="host.docker.internal:9464", job="grpc_eio", method="/example.Echo/Echo"}1

+ Add query

Standard (50c / 10k)

```
source: lib/grpc-eio-next/tmp/ghz_standard.txt
```

```
Summary:
  Count:      10000
  Total:      486.65 ms
  Slowest:    5.73 ms
  Fastest:    0.24 ms
  Average:    1.98 ms
  Requests/sec: 20548.57
```

Response time histogram:

Figure 1: Histogram of the number of steps per minute.

Latency distribution:

10 % in 1.24 ms
25 % in 1.57 ms
50 % in 1.94 ms
75 % in 2.34 ms
90 % in 2.74 ms
95 % in 2.98 ms
99 % in 3.69 ms

Status code distribution:

```
[OK] 10000 responses
```

High Concurrency (200c / 50k)

```
source: lib/grpc-eio-next/tmp/qhz_high.txt
```

```
Summary:
Count:      50000
Total:      2.57 s
Slowest:    25.45 ms
Fastest:    0.28 ms
Average:    8.81 ms
Requests/sec: 19467.05
```

Response time histogram:

Latency distribution:

```

10 % in 6.31 ms
25 % in 7.40 ms
50 % in 8.57 ms
75 % in 9.99 ms
90 % in 11.58 ms
95 % in 12.93 ms
99 % in 17.57 ms

```

Status code distribution:

```
[OK] 50000 responses
```

Interop Results – Raw Logs

Interop Suite (Go/Rust) – Repeat 1

source: lib/grpc-eio-next/tmp/interop_suite_fix_20260117_015547.log

```
Compiling echo-server v0.1.0 (/Users/dancer/me/.worktrees/grpc-fullspec/lib/grpc-eio-next/bench/rust-comparison)
Finished `release` profile [optimized] target(s) in 16.95s
Interop suite completed. Logs in /Users/dancer/me/.worktrees/grpc-fullspec/lib/grpc-eio-next/bench
```

Official Interop (plaintext)

source: lib/grpc-eio-next/tmp/interop_official_fix_plain_20260117_015500/run.log

```
==> Building interop_official_server (profile=release)
==> Starting interop_official_server on 127.0.0.1:50051
==> Running supported interop cases
PASS: empty_unary
PASS: large_unary
PASS: client_streaming
PASS: server_streaming
PASS: ping_pong
PASS: empty_stream
PASS: timeout_on_sleeping_server
PASS: cancel_after_begin
PASS: cancel_after_first_response
PASS: unimplemented_method
PASS: unimplemented_service
==> Skipped cases (unsupported): custom_metadata status_code_and_message special_status_message orca_per_rpc orca_oob pick_first_unary compute_en
==> Stopping server
Interop run completed.
```

Official Interop (TLS)

source: lib/grpc-eio-next/tmp/interop_official_fix_tls_20260117_015523/run.log

```
==> Building interop_official_server (profile=release)
==> Starting interop_official_server on 127.0.0.1:50051
==> Running supported interop cases
PASS: empty_unary
PASS: large_unary
PASS: client_streaming
PASS: server_streaming
PASS: ping_pong
PASS: empty_stream
PASS: timeout_on_sleeping_server
PASS: cancel_after_begin
PASS: cancel_after_first_response
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PASS: unimplemented_service
==> Skipped cases (unsupported): custom_metadata status_code_and_message special_status_message orca_per_rpc orca_oob pick_first_unary compute_en
==> Stopping server
Interop run completed.
```

4) Performance Baseline (ghz, localhost)

Standard (50c / 10k):

- RPS: 20,548.57
- P50: 1.94 ms, P95: 2.98 ms, P99: 3.69 ms
- tmp/ghz_standard.txt

High (200c / 50k):

- RPS: 19,467.05
- P50: 8.57 ms, P95: 12.93 ms, P99: 17.57 ms
- tmp/ghz_high.txt

Regression thresholds:

- docs/PERF-BASELINE.md

5) 24h Soak (low load)

Command:

```
CONCURRENCY=10 DURATION_PER_RUN=10s DURATION_SECONDS=86400 \  
./ops/soak/run_soak.sh
```

Window:

- Start: 2026-01-16 01:31:06
- End: 2026-01-17 01:31:04

Totals:

- OK: 682,799,327
- Unavailable: 168,977
- Canceled: 845
- Error rate: 0.024865%

Logs:

- tmp/soak_client.log
- tmp/soak_server.log
- tmp/soak_runner.log

Note:

- Unavailable errors are primarily expected at connection close boundaries.

Evidence Bundle

Logs:

- tmp/interop_suite_fix_20260117_015547.log
- tmp/interop_official_fix_plain_20260117_015500/run.log
- tmp/interop_official_fix_tls_20260117_015523/run.log
- tmp/prom_targets_verify.json
- tmp/prom_query_calls_verify.json
- tmp/ghz_standard.txt
- tmp/ghz_high.txt
- tmp/soak_client.log
- tmp/soak_server.log
- tmp/soak_runner.log

Screenshots:

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- docs/assets/prom_alerts.png

- docs/assets/prom_query_calls.png
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Load time: 11ms Result series: 2

grpc_server_calls_total{instance="host.docker.internal:9464", job="grpc_eio"}1

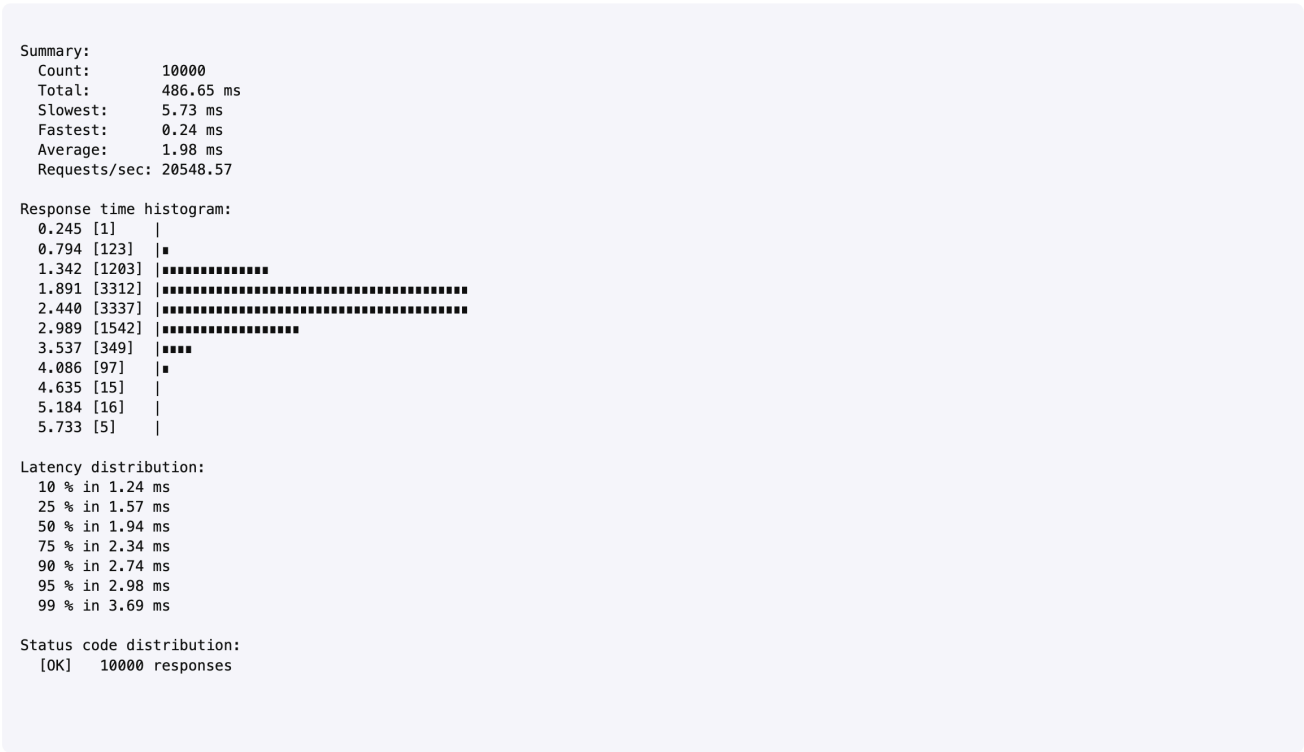
grpc_server_calls_total{instance="host.docker.internal:9464", job="grpc_eio", method="/example.Echo/Echo"}1

+ Add query

ghz Results (grpc-eio) - Raw Logs

Standard (50c / 10k)

source: lib/grpc-eio-next/tmp/ghz_standard.txt



High Concurrency (200c / 50k)

source: lib/grpc-eio-next/tmp/ghz_high.txt



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Official Interop (TLS)

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```
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==> Starting interop_official_server on 127.0.0.1:50051
==> Running supported interop cases
PASS: empty_unary
PASS: large_unary
PASS: client_streaming
PASS: server_streaming
PASS: ping_pong
PASS: empty_stream
PASS: timeout_on_sleeping_server
PASS: cancel_after_begin
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==> Skipped cases (unsupported): custom_metadata status_code_and_message special_status_message orca_per_rpc orca_oob pick_first_unary compute_en
==> Stopping server
Interop run completed.
```

Notable Fix During Validation

- Guarded response start on closed streams to avoid h2 assertion during timeout/cancel paths.